Practice 3: RF to EC conversion

Santiago Rodríguez Villabona - 2220387
Juan José De la Rosa Medina - 2202810

https://github.com/santiagoUIS/GNURADIO_COMUIS_2025_2_B1_G7/tree/Practica_3

School of Electrical, Electronic, and Telecommunications Engineering
Industrial University of Santander

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***Abstract*—A radio frequency (RF) signal is a high-frequency electromagnetic waveform used to transmit information through modulation techniques. However, analyzing such signals directly can be complex due to their rapidly varying carrier components. To address this, the concept of the complex envelope (CE) representation is introduced, which expresses the RF signal in a baseband form that captures its amplitude and phase variations without the high-frequency carrier[1].**

This report presents the process of converting an RF signal into its complex envelope representation using GNURadio. The main goal was to understand how this transformation simplifies the analysis and processing of digitally modulated signals, particularly those using OOK, BPSK, and FSK schemes. Throughout the study, several flowgraphs were designed and modified to observe the behavior of signals in the time, frequency, and constellation domains[2]. The results showed that working with the complex envelope not only makes the mathematical and spectral interpretation of signals easier but also provides a clearer visualization of modulation phenomena.

keywords: Complex envelope, radio frequency, OOK, BPSK, FSK, digital modulation, signal processing

I INTRODUCTION

In modern communication systems, representing a signal through its complex envelope (CE) is essential for simplifying the analysis and design of modulation and demodulation processes. Rather than working directly with high-frequency radio signals, the CE provides a compact and intuitive way to describe how amplitude and phase vary over time.

This study focuses on transforming radio frequency (RF) signals into their complex envelope representations using GNU Radio. By generating and analyzing different types of digital modulation, it becomes possible to examine how their characteristics appear in both the time and frequency domains.

The experimental procedure was developed to demonstrate how an RF signal can be represented and analyzed through its complex envelope form. Each stage of the experiment emphasizes the relationship between amplitude, phase, and frequency variations, and their corresponding effects on the envelope representation.

The first stage involved generating an On–Off Keying (OOK) signal, chosen for its simplicity and suitability for comparing the amplitude-modulated RF waveform with its CE equivalent. Observations in the time domain showed how the envelope reflects binary transitions between high and low amplitude levels, confirming that the CE efficiently captures the transmitted information while removing the high-frequency carrier component.

Understanding the relationship between RF and CE signals deepens the grasp of communication system fundamentals and enhances the ability to design efficient signal processing systems. Implementing these concepts in GNURadio provides a practical and interactive environment for visualizing modulations, effectively bridging the gap between theoretical understanding and real-world applications.

II METHODOLOGY

This section presents in an orderly and detailed manner the experiments carried out during the laboratory, including simulations and other considerations addressed within the practice.

II-A OOK modulation

On–Off Keying (OOK) modulation uses unipolar signals because its principle is based on representing digital bits by the presence or absence of a carrier. A “1” bit is transmitted by activating the carrier signal, while a “0” bit is represented by turning it off, without transmitting energy. This causes the resulting signal to have only positive or zero values, without inverting its polarity. Based on the diagram in Fig. 1, the signal is analyzed in its RF and EC form, throughout the time domain, and in the same way, the respective analysis of the signal in the frequency domain is performed.

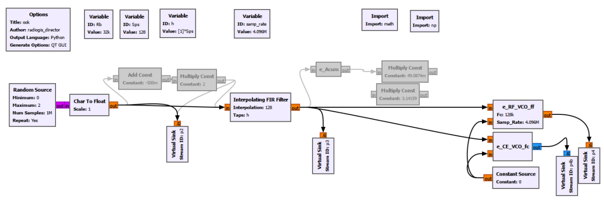

Figure 1. Block diagram for OOK modulation for a unipolar signal.

This step was crucial to understanding how the spectral distribution differs between the different versions of the modulated signal. While the RF spectrum showed sidebands around the carrier frequency, the CE spectrum remained centered, revealing the baseband nature of the complex envelope, as seen in Fig. 2.

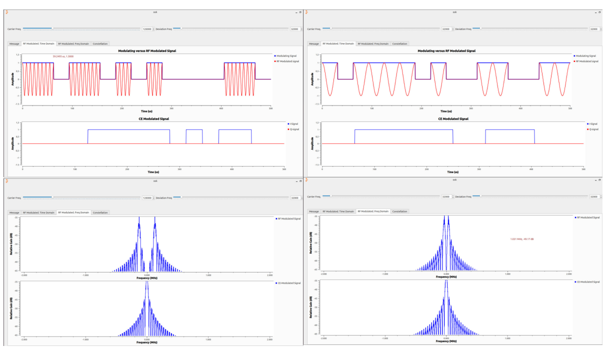

Figure 2. Unipolar signal modulated with OOK in its complex envelope and radio frequency version.

II-B BPSK modulation

Once amplitude-based modulation had been analyzed, the next stage focused on binary phase shift keying (BPSK) modulation. This modulation introduced phase changes instead of amplitude variations, allowing us to study how information can be transmitted by phase inversion, as shown in Fig. 4. The RF representation showed alternating phase states, while the CE version allowed for a much clearer visualization of these transitions, as the amplitude remained constant but the phase rotated between two distinct positions. The block diagram created to implement BPSK modulation can be seen in Fig. 3.

II-C FSK modulation

To extend the analysis, Frequency Shift Keying (FSK) modulation was implemented with the block diagram of the Fig. 5. In this case, the focus was placed on how changes in the instantaneous frequency of the carrier

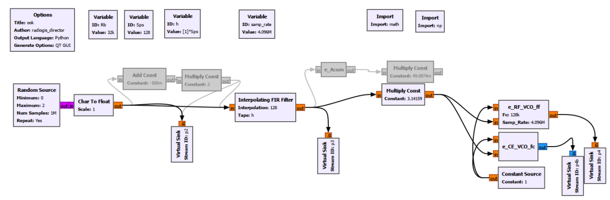

Figure 3. Block diagram for BPSK modulation for a unipolar signal.

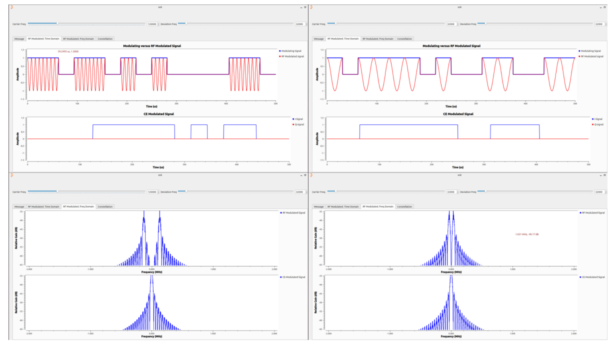

Figure 4. Bipolar signal modulated with BPSK in its complex envelope and radio frequency version.

translate into the CE domain. Two distinct frequencies were assigned to represent binary symbols ‘0’ and ‘1’. Observations showed that, while the RF signal alternated between two tones around a central carrier, the CE signal exhibited smooth frequency transitions that were easier to track and interpret, as shown in Fig. 6.

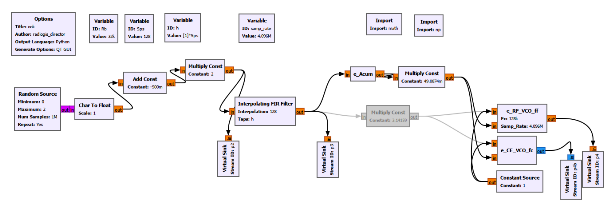

Figure 5. Block diagram for FSK modulation for a bipolar signal.

Subsequently, the frequency-domain analysis for FSK was carried out. By modifying both the carrier frequency and the deviation parameter, it was possible to identify how the spacing between tones affected the clarity of the modulated signal. A larger deviation produced more separation between the frequency components, reducing overlap in the spectrum and improving distinguishability.

Finally, the constellations for each modulation type were plotted to visualize the signal states in a two-dimensional plane, as shown in Fig. 7. The OOK constellation showed two amplitude points, the BPSK displayed

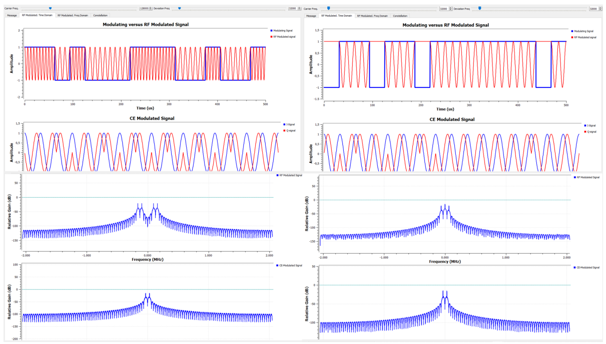

Figure 6. Bipolar signal modulated with FSK in its complex envelope and radio frequency version.

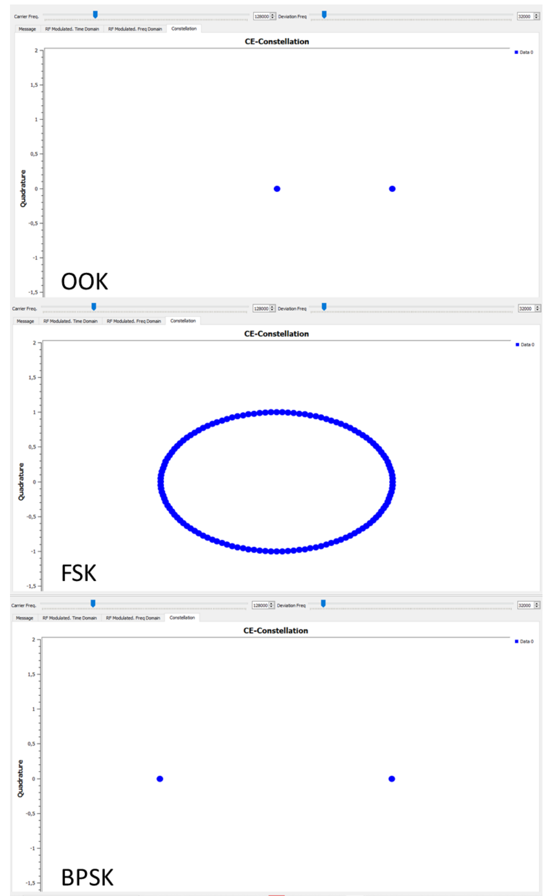

Figure 7. Constellation diagram for the different proposed modulations.

two opposite phase positions on the same radius, and the FSK exhibited multiple clusters corresponding to different frequency states. This graphical interpretation provided a direct understanding of how each modulation type encodes information differently.

All data obtained from the experiment were recorded and compared to theoretical expectations. The analysis confirmed that the CE representation simplifies the visualization and interpretation of modulated signals by eliminating the need to process the high-frequency components directly. The entire methodology ensured that each modulation was correctly implemented, analyzed, and validated under the same conditions for consistent comparison.

III EXPERIMENTS AND RESULTS

This section presents the experimental analysis and results obtained from the RF to Complex Envelope (CE) conversion for the OOK, BPSK, and FSK modulation schemes. Each experiment highlights the differences between the RF representation and its corresponding CE form in both time and frequency domains, as well as their constellations.

III-A OOK Modulation Analysis

The OOK signal was generated using a unipolar digital input to control the presence or absence of the carrier. In the time domain, the RF signal showed a clear carrier burst when the bit “1” was transmitted, and complete suppression when transmitting “0”. The complex envelope, on the other hand, retained only the amplitude variations, eliminating the high-frequency oscillations and showing the binary pattern directly.

In the frequency domain, the RF version displayed two symmetrical sidebands around the carrier frequency, while the CE remained centered at baseband. This confirmed that the spectral information of the signal can be fully represented without the carrier component.

III-B BPSK Modulation Analysis

The BPSK signal was generated by applying phase inversion of 180° to represent binary symbols. In the RF representation, phase transitions appeared as sudden changes in waveform polarity, while in the CE, the information was clearly observed as rotations between two opposite points in the complex plane.

The spectrum for BPSK showed a sinc-shaped main lobe with symmetrical side lobes. In CE, this spectral shape was preserved but centered at zero frequency, highlighting again that the CE captures the same information as RF without including the carrier term.

III-C FSK Modulation Analysis

The FSK modulation was implemented by assigning two distinct carrier frequencies to binary symbols "0" and "1". In the time domain, the RF signal alternated between two sinusoidal tones, while the CE version showed smooth frequency transitions between two baseband values corresponding to those tones.

In the frequency domain, varying the frequency deviation parameter allowed controlling the spacing between tones. Larger deviations resulted in clearer separation between the spectral peaks, minimizing interference between symbols.

III-D Constellation Diagrams

Finally, constellation diagrams were plotted to represent the symbol mapping for each modulation. In OOK, two points along the amplitude axis represent "on" and "off" states. In BPSK, two points appear on opposite sides of the in-phase axis, corresponding to phase shifts of 0° and 180°. For FSK, the constellation forms clusters corresponding to different frequency states, representing the transitions between tones.

Overall, the results demonstrated that the complex envelope accurately represents the essential information of the modulated signal, preserving its amplitude, phase, and frequency variations while simplifying its visualization and spectral analysis.

IV CONTRIBUTIONS

Throughout this study, several key concepts and relationships between radio frequency (RF) and complex envelope (CE) signals were reinforced. The analysis contributed to a better understanding of how information is encoded and preserved across different digital modulation schemes and how their characteristics manifest in both time and frequency domains.

The main contributions of this laboratory are summarized as follows:

- Strengthened comprehension of the mathematical relationship between an RF signal and its CE representation, emphasizing how amplitude, phase, and frequency components are mapped to baseband.
- Identification of the distinctive behaviors of OOK, BPSK, and FSK modulations, highlighting how each encodes information through variations in amplitude, phase, or frequency.
- Recognition that the carrier frequency only shifts the RF spectrum, while the CE remains invariant and centered at baseband.
- Validation that the complex envelope simplifies visualization and analysis by isolating the information-bearing components of the signal.

These insights provide a theoretical foundation for the study of digital communication systems, allowing a more intuitive and efficient understanding of modulation processes and their spectral implications.

V CONCLUSIONS

The analysis of the conversion from radio frequency (RF) to complex envelope (CE) representation allowed for a clear understanding of how the essential information of a digitally modulated signal can be preserved while removing the carrier component. The CE form retains the amplitude, phase, and frequency variations that define the modulation, but simplifies their observation and interpretation by representing them at baseband.

For OOK modulation, it was confirmed that the CE accurately reflects the amplitude changes corresponding to binary data, maintaining the same information content as the RF signal. In BPSK, the phase inversion between symbols was easily identifiable through the CE representation, highlighting its effectiveness for visualizing phase-based modulations. For FSK, the CE provided a direct view of the instantaneous frequency transitions, demonstrating how the spacing between tones affects signal distinction and spectral separation.

It was also verified that changes in the carrier frequency only shift the RF spectrum but do not affect the CE representation, which remains centered at baseband. In contrast, modulation parameters such as amplitude, phase shift, and frequency deviation directly influence the signal's shape and constellation distribution.

In summary, the study confirmed that representing a signal through its complex envelope simplifies analysis and provides a clearer insight into how digital modulation schemes encode information through amplitude, phase, or frequency variations.

References

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